

HARSHITHA PUTTASWAMY

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CERTIFICATIONS

AWS Certified Machine Learning Engineer - Associate

PROFESSIONAL EXPERIENCE

- Research Assistant, Scalable LLM Inference Framework for Fairness Evaluation** *February 2025 – Present*
Binghamton University, Department of Computer Science *Binghamton, NY, USA*
- Developed a scalable inference framework using large Language Models (LLMs) to evaluate the envy-freeness in item allocations for n agents, enabling consistent benchmarking across multiple allocation algorithms.
 - Increased inference throughput by four times by orchestrating multi-threaded LLM execution via distributed Ollama instances across multiple GPUs, accelerating real-time evaluation of fairness properties.
 - Designed prompt engineering strategies to elicit agent preferences, integrating statistical tests to analyze independence between algorithms and personality-driven behavior, validated against MTurk-collected human responses.
- Research Assistant, Reinforcement Learning & Algorithmic Optimization** *January 2024 – December 2024*
Binghamton University, Department of Computer Science *Binghamton, NY, USA*
- Developed and implemented a dynamic framework using Multi-Armed Bandit (MAB), a Reinforcement Learning algorithm to optimize iterative voting decisions, improving collective utility across agents, and modeling sequential decision-making similar to ad platform bidding strategies.
 - Engineered scalable algorithms using Integer Linear Programming (ILP) to solve multi-agent resource allocation and ranking problems, enabling strategic budget distribution across constrained environments.
 - Conducted extensive experimental analysis on preference aggregation and rule convergence, producing insights on maximizing utility and satisfaction, applicable to auction mechanisms and real-time bidding environments.
- Backend Developer, Internship** *March 2022 – December 2022*
Fyle Technologies Pvt Ltd *Bangalore, India*
- Developed and optimized RESTful APIs using Django, Angular, and PostgreSQL (PSQL), contributing to 8 new features that increased market competitiveness and user engagement.
 - Enhanced code reliability by introducing Pytest-based unit/integration tests and implementing GitHub Actions for CI, achieving 98%+ test coverage and reducing code review time by 90%.

PROJECT EXPERIENCE

- Immigration Law Chatbot (Fine-Tuned LLM with RAG on AWS) | AWS, Mistral-7B, RAG** *May 2025 – June 2025*
- Generated a custom legal Q&A dataset, comprising over 9,000 pairs, by efficiently processing the Immigration and Nationality Act PDF and leveraging advanced prompt engineering with the Qwen2.4 14B model.
 - Fine-tuned Mistral-7B-instruct model to optimize the LLM's performance for legal Q&A using AWS SageMaker training jobs via the Hugging Face SDK.
 - Reduced hallucination rate and improved answer relevance by implementing a Retrieval-Augmented Generation (RAG) system, integrating relevant legal documents from AWS S3 into a vector database built with Amazon OpenSearch.
 - Deployed the fine-tuned LLM onto a SageMaker endpoint, enabling scalable and accessible inference via a robust serverless architecture implemented with AWS Lambda and API Gateway.
 - Established efficient data pipelines for both LLM training and RAG knowledge base by reliably uploading and retrieving large datasets and source documents from AWS S3.
- Splitwiser – A Splitwise clone | Full Stack Development, Amazon Bedrock** *June 2024 – December 2024*
- Created a Splitwise-inspired application featuring receipt scanning via camera or file upload, group and non-group expense tracking, and dynamic graphical and manual split functionalities.
 - Leveraged Amazon Bedrock API to convert receipt images into JSON data, facilitating graphical expense visualization.
 - Utilized NextJS for UI and API development, Google Firebase for backend services, and Firestore for storing receipt data.
- Autokube – Kubernetes Cluster Automation Platform | DevOps, CI/CD** *August 2021 – March 2022*
- Led the development of the web application for automating Kubernetes cluster creation and modification.
 - Utilized MongoDB, Express.js, React, Node.js (MERN stack) for formulating APIs to initiate scripts, triggering a series of workflows for managing the backend Kubernetes clusters.

EDUCATION

Binghamton University, State University of New York *Binghamton, NY, USA*
Master of Science in Computer Science *January 2023 – December 2024*

TECHNICAL SKILLS

Programming Languages and DB: C, Python, Java, JavaScript, TypeScript, MongoDB, PostgreSQL, MySQL, Firebase
Frameworks and Software: Amazon SageMaker, Amazon Bedrock, AWS Lambda, Amazon OpenSearch, DynamoDB, EC2, ExpressJS, ReactJS, NodeJS, NextJS, Django, CI/CD, Git, Linux, Kubernetes, TensorFlow, Keras, PyTorch, Pandas, Google Colab